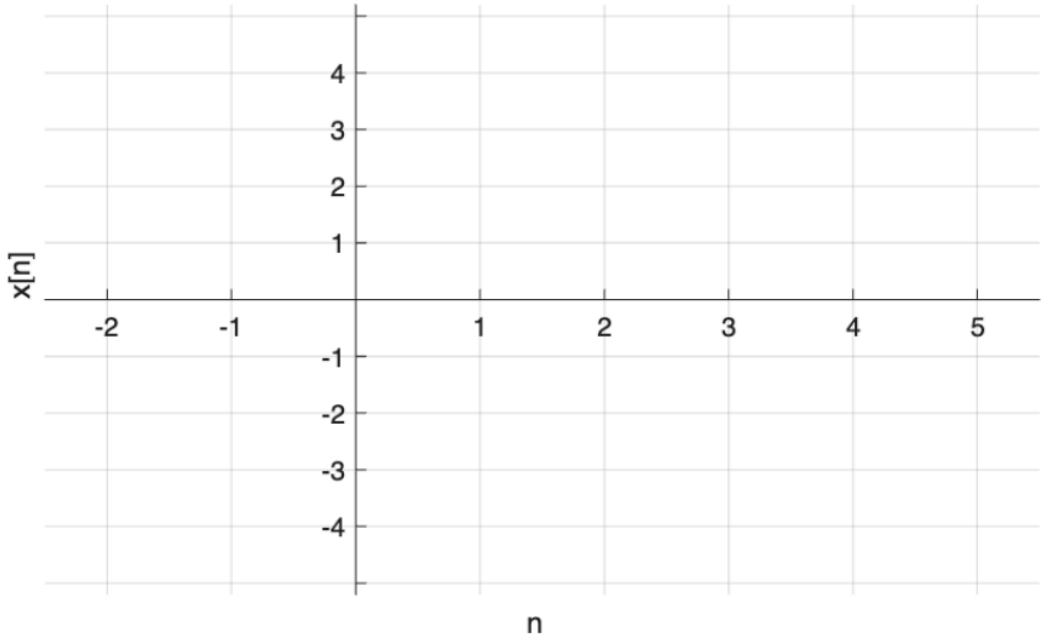
Sampling, DT signals, system properties

Practice questions · Day 02 · Sun 25 Oct · pack D1

5. Practice

From your course files

Sample Test 1 Q1 (sample test, 5 marks). Sampling.
(a) The continuous time signal $x_a(t) = 10e^{-0.01t}\cos(400\pi t)$ (t in seconds) is sampled at $F_s = 100$ Hz. Determine the sampling interval T_s and evaluate the samples of $x(n)$ for $0 \le n \le 3$. [3]
(b) On the axes provided, sketch the stem plot of the DT signal $x(n) = 2^{-n}\big(u(n+1) - u(n-2)\big) + \delta(n-3)$. [2]



Sample Test 1 Q1(b): the axes provided on the test paper

Sample Test 1 Q2 (sample test, 4 marks). DT Signal Operations. Consider the DT signal $x(n) = \{8,\ 9,\ \underline{1},\ 9,\ 6,\ 1\}$ (the underline marks $n = 0$). Samples at all other indices n are zero.
(a) Write in sequential form the signal $x(n-3) + x(n+1)$. [1]
(b) Write in sequential form the signal $x(\frac{1}{2}n)$, i.e. upsampled by a factor of 2. [1]
(c) Write in sequential form the signal $x(n)(-1)^n$. [1]
(d) Calculate the energy of $x(n)$. [1]

Sample Test 1 Q3 (sample test, 4 marks). DT System Properties. For each of the following systems, tick all the boxes that apply: ☐ Linear ☐ Time invariant ☐ Causal ☐ Static/memoryless.

(a) $y[n] - ny[n-1] = x[n] + x[n-1]$. [1]

(b) $y[n] - y[n-1] + y[n-2] = \cos(0.2\pi)x[n]$. [1]

(c) $y[n] - 0.1y[n-1] = x[n+1] - 1$. [1]

(d) $y[n] - y[n-1] - y[n-4] = x[n] + x[n+1]$. [1]

T1 Q1 (tutorial). Determine whether or not each of the following signals is periodic. In case a signal is periodic, specify its fundamental period.

(a) $x(t) = 3\cos(5t + \pi/6)$

(b) $x(n) = 3\cos(5n + \pi/6)$

(c) $x(n) = 2\exp\left(j(n/6 - \pi)\right)$

(d) $x(n) = \cos(n/8)\cos(\pi n/8)$

(e) $x(n) = \cos(\pi n/2) - \sin(\pi n/8) + 3\cos(\pi n/4 + \pi/3)$

T1 Q2 (Nyquist rate, DT frequencies and ideal D/A output for $x(t) = \sin(480\pi t) + 3\sin(720\pi t)$ sampled at 600 Hz) is worked in full as Example 1 in Section 3. Cover the working and redo it: 720 Hz; $\omega = 0.8\pi$ rad; $y(t) = -2\sin(480\pi t)$.

T1 Q3 (tutorial). The discrete time signal $x(n) = 6.35\cos(\pi n/10)$ is quantised with a resolution (a) $\Delta = 0.1$ or (b) $\Delta = 0.02$. How many bits are required in the A/D converter in each case?

T1 Q4 (tutorial). Determine the bit rate and the resolution in the sampling of a seismic signal with dynamic range of 1 Volt if the sampling rate is $F_s = 20$ samples/s and we use an 8-bit A/D converter. What is the maximum seismic signal frequency that can be recovered from the resulting digital signal without distortion?

T2 Q1 (tutorial). Determine whether or not each of the following systems is static, linear, time-invariant, causal and stable.

(a) $y(n) = \cos\big(x(n)\big)$
(b) $y(n) = \sum_{k=-\infty}^{n+1} x(k)$
(c) $y(n) = x(n)\cos(\omega_0 n)$
(d) $y(n) = x(-n+2)$
(e) $y(n) = |x(n)|$
(f) $y(n) = x(2n)$

T2 Q2 (tutorial). The following input-output pairs have been observed during the operation of a *time-invariant* system T:

$$x_1(n) = \{\underset{\uparrow}{1}, 0, 2\} \xrightarrow{T} y_1(n) = \{\underset{\uparrow}{0}, 1, 2\}$$

$$x_2(n) = \{\underset{\uparrow}{0}, 0, 3\} \xrightarrow{T} y_2(n) = \{\underset{\uparrow}{0}, 1, 0, 2\}$$

$$x_3(n) = \{\underset{\uparrow}{0}, 0, 0, 1\} \xrightarrow{T} y_3(n) = \{1, \underset{\uparrow}{2}, 1\}$$

Can you draw any conclusions regarding the *linearity* of the system?

New questions

N1 (easy) [3] $x(t) = 4\cos(300\pi t) + 2\sin(900\pi t)$ is sampled at $F_s = 500$ Hz. (a) Give the Nyquist rate. [1] (b) Find $x(n)$ with all frequencies in $(-\pi, \pi]$. [1] (c) Find the ideal DAC output $y(t)$. [1]

N2 (easy) [2] (a) Find the fundamental period of $x(n) = \cos(0.3\pi n) + \sin(0.45\pi n) + e^{j2\pi n/5}$. [1.5] (b) Is $\cos(0.3n)$ periodic? [0.5]

N3 (medium) [3] A sensor signal spans -2.5 V to $+2.5$ V and is sampled at 8 kHz. The resolution must be at most 1 mV. (a) Find the minimum number of bits. [1] (b) Find the actual Δ and the worst-case rounding error. [1] (c) Find the bit rate and the highest frequency that can be recovered. [1]

N4 (medium, chained) [4] $x(n) = \{2, -1, \underset{\uparrow}{3}, 1, 4\}$. (a) Find $x(n-2) - x(n+1)$. [1] (b) Find $x(-2n+1)$. [1] (c) Find the even part $x_e(n) = \frac{1}{2}[x(n) + x(-n)]$. [1] (d) Find E_x and say whether x is an energy or a power signal. [1]

N5 (medium) [3] Classify each system (static, linear, time-invariant, causal, stable), giving a counterexample for each property that fails. (a) $y(n) = e^{x(n)}$ (b) $y(n) = \sum_{k=n-2}^{n+2} x(k)$ (c) $y(n) = x(n^2)$

N6 (hard, chained) [5] A radar transmits $x(n) = \{\underset{\uparrow}{1}, 2, -1\}$. The received echo is $y(n) = 0.5\,x(n - D)$. Use the L03 definition $r_{yx}(l) = \sum_n y(n)x(n - l)$. (a) Find $r_{xx}(l)$ for all l. [1] (b) Take $D = 3$. Write $y(n)$ and compute $r_{yx}(l)$. [2] (c) Find $\hat{d}$ and show that $r_{yx}(l) = 0.5\,r_{xx}(l - D)$. [1] (d) Take $F_s = 1$ MHz and $c = 3 \times 10^8$ m/s. Estimate the range r. [1]

6. Quick recall check

1. Is $\cos(0.7n)$ periodic?
2. With $F_s = 1$ kHz, at what frequency does a 1.3 kHz cosine appear after ideal reconstruction?
3. How many bits does a 300-level quantiser need?
4. Is $y(n) = x(n) + 2$ linear? Is it time-invariant?
5. What is $r_{xx}(0)$ equal to?